FLORIA TSUI

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EDUCATION

BACHELOR of SCIENCE: Computer Science & Political Science

Brown University | 2019-2023 | GPA: 3.80

Past courses: Fundamentals of Computer Systems, Object Oriented Programming, Functional Programming, Algorithms and Data Structures

Current courses: Deep Learning, Software Security and Exploitation

EXPERIENCE

TEACHING ASSISTANT

Introduction to Computer Systems | Fall 2020
Computing Foundations: Data | Spring 2020

- 1 of 14 TAs hired as a freshman out of 100+ applicants
- Graded homework assignments for 150+ students and held office hours, helping students debug code in Pyret and Python (4 hours / week)
- Developed and implemented 3 course projects: analyzing large datasets, building a maze game using lists in Pyret, and generating a book recommendation site using hashtables in Python

RESEARCH ASSISTANT

Department of Sociology | Summer 2020

- Read and flagged discussion of climate change in Presidential debate transcripts from 2000-2020 (45 in total)
- Developed a dictionary of technocratic terms and Python code to analyze the use of technocratic language in climate change debates between Presidential Candidates

SUMMER FIELD INTERN

Generation Rising and Everyday People PAC | Summer 2020

- Phone banked to Florida and Arizona, making over 300 calls in 2 weeks (9 hours / week)
- Textbanked to Pennsylvania, texting 3000+ people to register people to vote

FINANCE FELLOW

Gretchen Driskell for Congress | Summer 2020

- Directly assisted Finance Director with fundraising events by finding and analyzing prospective attendees, designing invitations, and aiding in recruitment
- Helped in administrative tasks including finding prospective donors for interns to research and analyzing other candidates' fundraising patterns
- Raised 200+ through my own fundraising efforts

PROJECTS

SHARDING

C++ | Spring 2020

- Created a sharded, distributed key-value storage system using gRPCs and protobufs
- Implemented a "shardmaster" that assigns shards to key-value servers and updates the shard distributions when servers join or leave
- Constructed a key-value server, where users can get, put, and append and transfer key-value pairs depending on the shard configuration

SEARCH

Scala | Spring 2020

- Wrote a search engine for users to query a corpus of Wikipedia pages and returns the most relevant pages
- Developed an Indexer that processes the text of each page that stores each word and its frequency into hashtables, using Google's page rank algorithm (< 1 min)
- Developed a Querier that processes a free text query from the user, calculates the relevance score, and returns the pages associated with the query (< 1 sec)

WEENSYOS

C++ | Spring 2020

- Separated the address space of the operating system for kernel and process isolation
- Implemented the basics of virtual to physical memory address architecture using pagetables
- Wrote the system calls fork and exit

VUNMO

C++ | Spring 2020

- Implemented a concurrent banking service, similar to the application Venmo
- Created a synchronized queue using condition variables for a pool of "worker" threads that would handle user requests
- Developed a multi-threaded server that accepts clients, recieve and read client requests, and process each request

SKILLS & INTERESTS

PROGRAMMING LANGUAGES- **C, Java, Python, Scala, C++**

DESIGN- **Adobe Photoshop, Illustrator, Canva**

- Staff Illustrator for the College Hill Independent, contributing 4 illustrations (5 hours / week)